Tracking Semantic Change in Technology-Related Words

Case study: Dynamic Embeddings for Temporal Language Change

Dataset: News Category Dataset (2012–2022)

Periods: 2010s (2012–2019) and 2020s (2020–2022)

Method: Word2Vec Skip-gram + Orthogonal Procrustes alignment

Target words: virus, platform, network, model, web, stream

Control words: water, city, winter, animal, family, school

Main idea:

Train separate embeddings for different periods, align them,

and measure semantic shift using cosine distance and nearest neighbors.

Results Summary

Main results (2020s cosine distance from base period):

- model: 0.541
- network: 0.535
- platform: 0.530
- virus: 0.346

Interpretation:

- virus shifted toward pandemic-related contexts (COVID-19).
- platform and network shifted toward digital/media/political contexts.
- model showed the strongest semantic shift in this experiment.

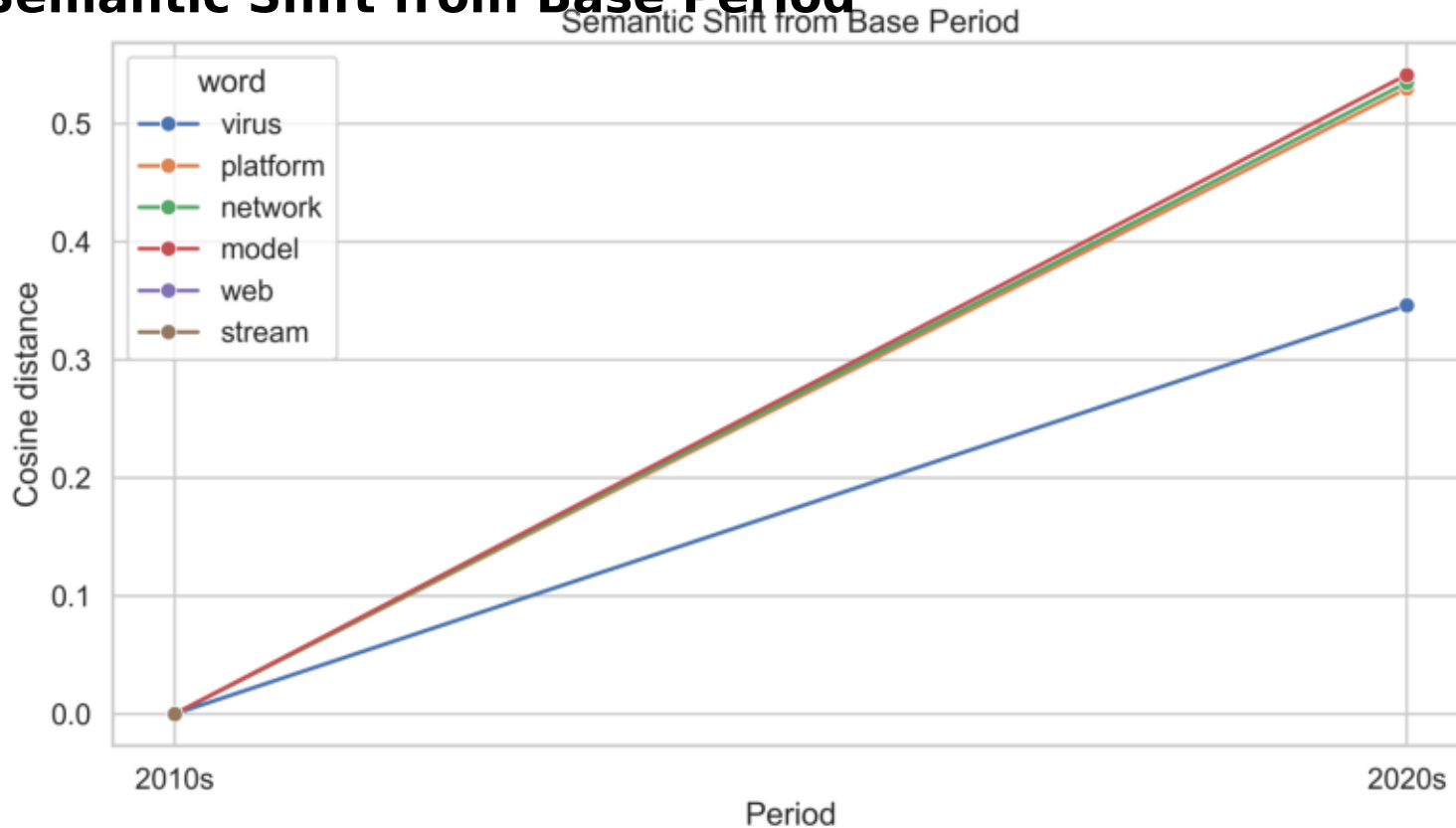
Group comparison:

- control: 0.494
- target: 0.488

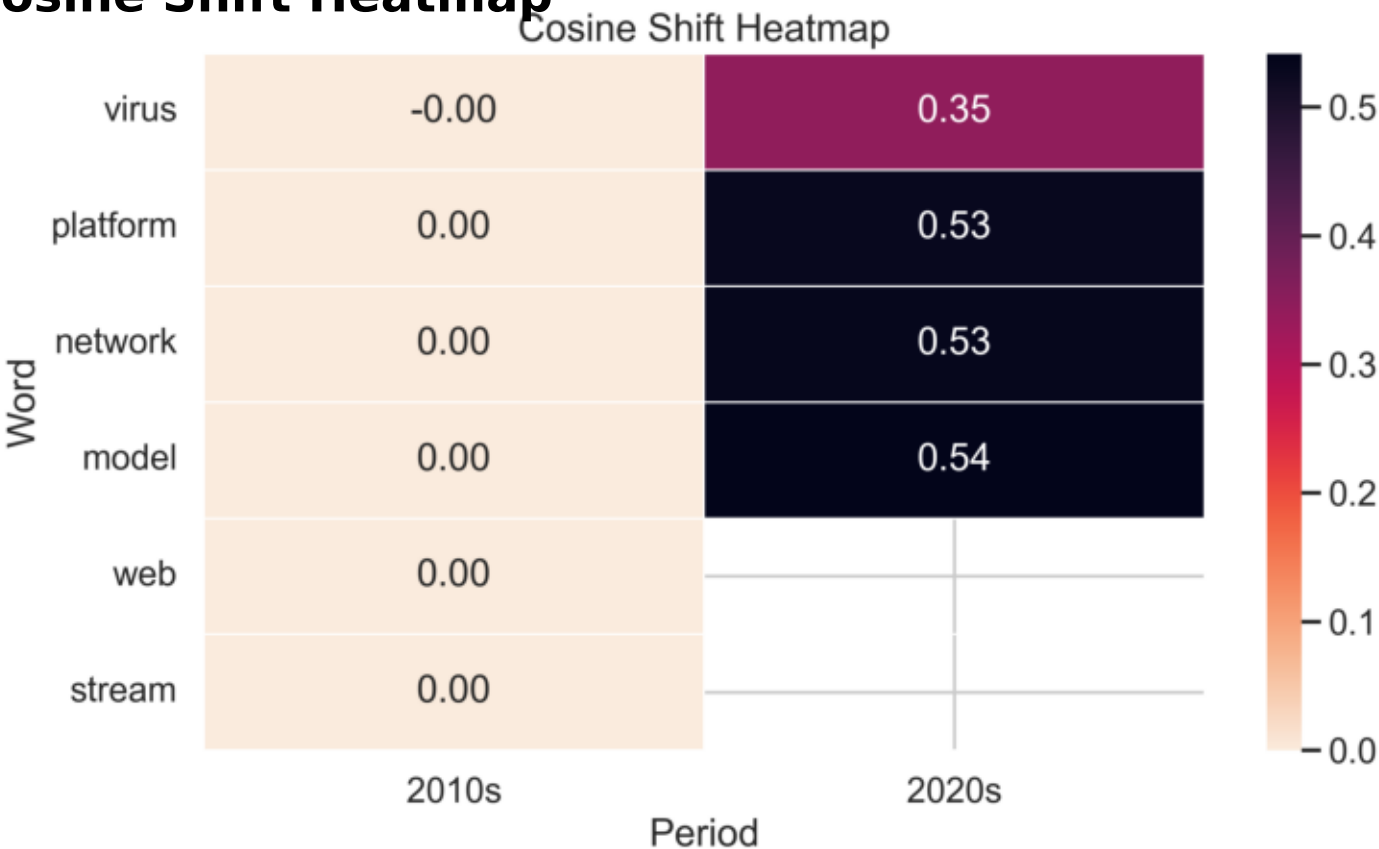
Conclusion:

Temporal embeddings captured meaningful semantic shifts for several words, although the target vs. control difference was modest.

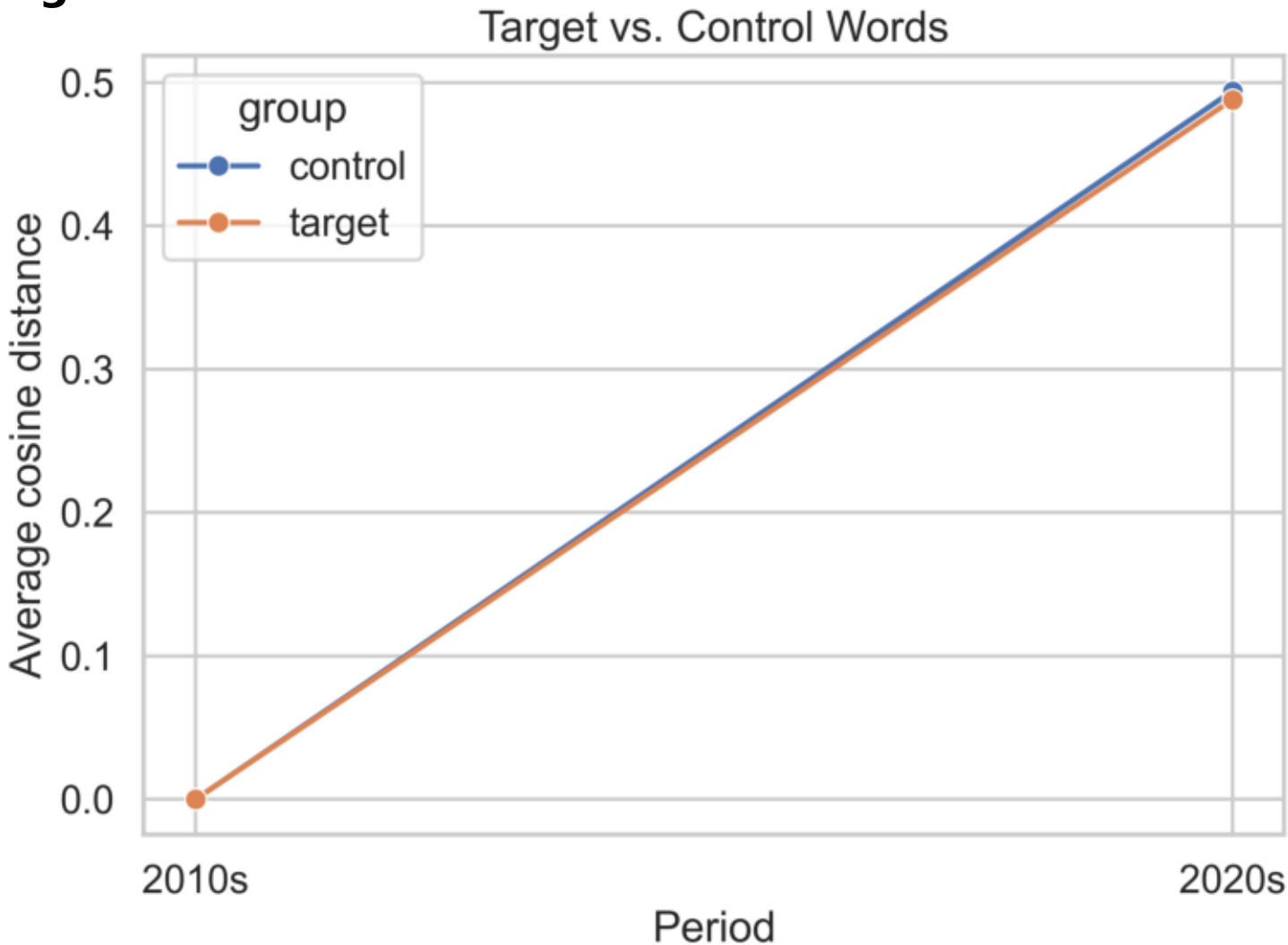
Semantic Shift from Base Period



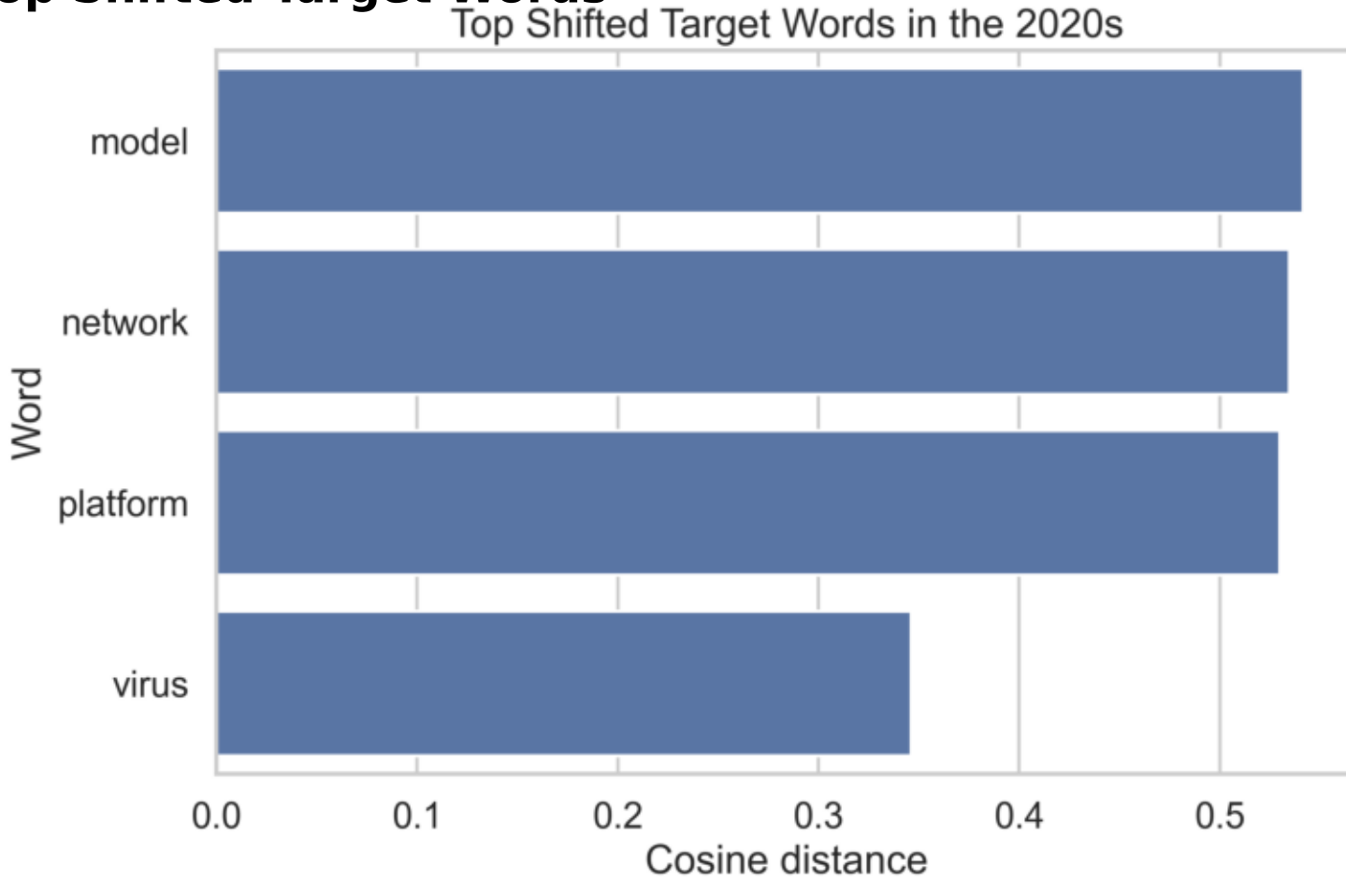
Cosine Shift Heatmap



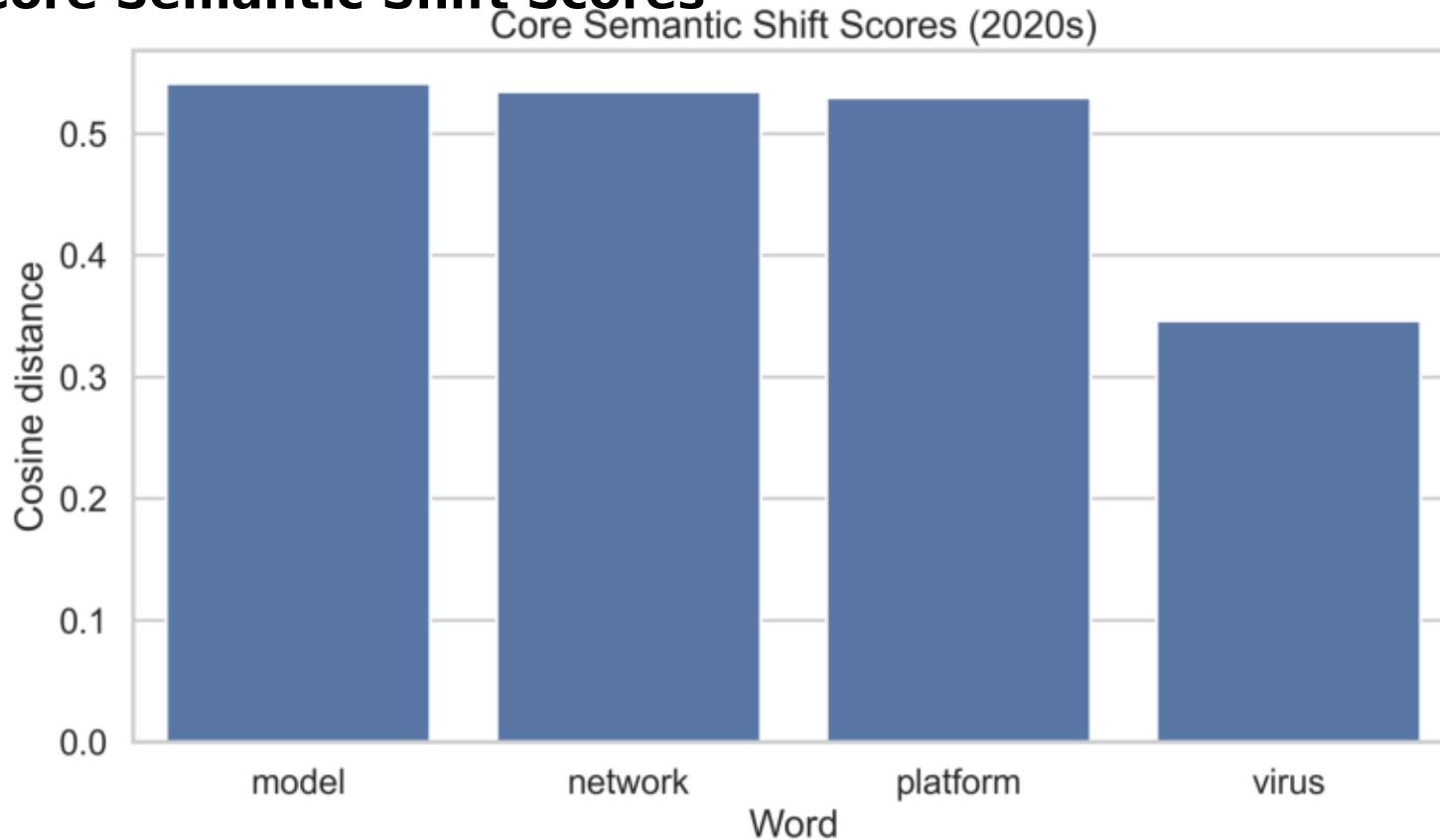
Target vs. Control Words



Top Shifted Target Words



Core Semantic Shift Scores



Nearest Neighbors Across Periods

	2010s	2020s
virus	mosquito-borne, zika, influenza, sars-like, vaccine, transmission, transmitted, outbreaks, drug-resistant, infections	disease, cases, outbreak, surge, low, infections, number, omicron, rise, china
platform	messaging, acquisitions, non-profit, broadcasting, guerrilla, left-wing, blockchain, aggregation, spanish-language, flexible	surprising, so-called, cracked, error, patriotic, influential, nsfw, greatest, midnight, drag
network	stylelist, homerun, trop, whydid, kirschner, keiko, ramshackle, beckerman, kailani, mclendon	hypocrisy, radio, exposes, personality, late-night, rant, bonkers, business, mocking, right-wing
model	models, lawley, supermodel, plus-size, img, daria, vodianova, narciso, mag, lingerie	stunning, heartbreaking, amazing, mogul, hadid, grammy, alongside, dennis, dubbed, career